

University of Colorado at Colorado Springs
Department of Mechanical and Aerospace Engineering

SPCE 5065

HOMEWORK #4

DUE: Friday 27 July 2026

Instructions:

- Include your references at the end of your homework in AIAA format.
<https://www.aiaa.org/publications/journals/reference-style-and-format>
- Include units in your final answers!
- Write non-mathematical answers in complete sentences.
- Clearly state your assumptions.

1. For the current events presentations this week

- a) Summarize the presentation
- b) Describe something you learned from it
- c) Write one question you have left about the presentation

2. In plasma physics, a Debye length is a measure of a charge carrier's net electrostatic effect in a plasma and how far its electrostatic effect persists.

- a. Determine the Debye length in the lower ionosphere at an altitude of 300 km, where the electron temperature is approximately 1500 K. Assume a daytime solar max plasma density.
- b. Determine the Debye length in the ionosphere at an altitude of 1000 km, where the electron temperature is approximately 5000 K. Assume a daytime solar max plasma density.

3. An electromagnetic signal traverses the ionosphere vertically along a path that has a total electron content of 10^{18} electrons/m².

- a. Determine the vertical time delay for a frequency of 150 MHz.
- b. Determine the excess range for 150 MHz if the vacuum speed of light were used.
- c. Determine the time delay if the frequency were 1.6 GHz.
- d. Determine the excess range for 1.6 GHz if the vacuum speed of light were used.

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4. Find an online ionospheric model. Provide an example of the model and describe it. Include in your description, at a minimum, who publishes it, where the data is gathered from, and any limitations of the model.

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5. Spacecraft in high altitude orbits are especially susceptible to spacecraft charging and can attain high potentials. Because the plasma is at a high temperature, the average speeds of the protons and ions are very large.

If we wish to minimize the induced current in a spherical geostationary satellite, how large will the spacecraft voltage be? The plasma temperature is 10^7 K.

Clearly state any assumptions you make.

The first order currents generated by the plasma environment for a spherical satellite are:

For electrons:

$$I_e = I_{e,o} A_e e^{\frac{eV}{k_B T_e}} \text{ for } V < 0 \text{ (repelled)}$$

$$I_e = I_{e,o} A_e \left[1 + \frac{eV}{k_B T_e} \right] \text{ for } V > 0 \text{ (attracted)}$$

Where:

$$I_{e,o} = \frac{1}{4} e n_e v_{mean}$$

$$v_{mean} = \sqrt{\frac{8k_B T_e}{\pi m_e}} - v_{spacecraft}$$

Where:

e = electron charge, C

V = voltage (V)

k_B = Boltzmann's constant, J/K

T_e = electron temperature, K

n_e = electron number density 1/m³

v_{mean} = relative velocity of electrons to spacecraft velocity (km/s)

For ions:

$$I_i = I_{i,o} A_i e^{\frac{eV}{k_B T_i}} \text{ for } V > 0 \text{ (repelled)}$$

$$I_i = I_{i,o} A_i \left[1 + \frac{eV}{k_B T_i} \right] \text{ for } V < 0 \text{ (attracted)}$$

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Where:

$$I_{i,o} = \frac{1}{4} e n_i v_{mean}$$

$$v_{mean} = \sqrt{\frac{8k_B T_i}{\pi m_i}} - v_{spacecraft}$$

Where:

T_i = ion temperature, K

n_i = ion number density $1/m^3$

v_{mean} = relative velocity of ions to spacecraft velocity (km/s)

- a. What is the speed of the spacecraft?
 - b. What are the mean speeds of the ions and electrons?
 - c. Find expressions for the induced potential currents for the electrons and ions in terms of V (spacecraft voltage). State your assumptions.
 - d. Find an expression in terms of V for total current flow.
 - e. For no current flow, what would the spacecraft voltage be?
 - f. Is this a high risk to the spacecraft? If so, what would you recommend to reduce the risk?
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6. One of the reasons most earth orbiting spacecraft are negatively biased is due to the more widespread use of npn versus pnp transistors.
 - a. What are npn and pnp transistors and what are their advantages and disadvantages?
 - b. Why might they be of wider use on earth orbiting spacecraft than pnp transistors?

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7. Select an article in a peer-reviewed journal or NASA document that describes the electrical grounding of spacecraft and summarize its contents.

8. Determine the voltage with respect to its environment of a spacecraft at synchronous altitude if the plasma temperature is 1×10^7 K. Consider the environment to consist primarily of electrons and protons. Clearly state your assumptions.